

```
# RAT - NUMBER - ROW
def rat_num_row():
    f=open('patterns.txt', 'w+')
    f.write("====RAT NUMBER ROW====\n")
    for i in range(1,6):
        for j in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
rat_num_row()

# RAT - NUMBER - COL
def rat_num_col():
    f=open('patterns.txt', 'a+')
    f.write("====RAT NUMBER COL====\n")
    for i in range(1,6):
        for j in range(0,i):
            f.write(str(j))
        f.write('\n')
    f.close()
rat_num_col()

# RAT - STAR
def rat_star():
    f=open('patterns.txt', 'a+')
    f.write("====RAT STAR====\n")
    for i in range(1,6):
        for j in range(0,i):
            f.write('*')
        f.write('\n')
    f.close()
rat_star()

# RAT - UPPER - ROW
def rat_upper_row():
    f=open('patterns.txt', 'a+')
    f.write("====RAT UPPER ROW====\n")
    for i in range(1,6):
        for j in range(0,i):
            f.write(chr(i+64))
        f.write('\n')
    f.close()
rat_upper_row()
```

```
def rat_upper_col():
    f=open('patterns.txt', 'a+')
    f.write("====RAT UPPER COL====\n")
    for i in range(1,7):
        for j in range(1,i):
            f.write(chr(j+64))
        f.write('\n')
    f.close()
rat_upper_col()

# RAT - LOWER - ROW
def rat_lower_row():
    f=open('patterns.txt', 'a+')
    f.write("====RAT LOWER ROW====\n")
    for i in range(1,6):
        for j in range(0,i):
            f.write(chr(i+64+32))
        f.write('\n')
    f.close()
rat_lower_row()

# RAT - LOWER - COL
def rat_lower_col():
    f=open('patterns.txt', 'a+')
    f.write("====RAT LOWER COL====\n")
    for i in range(1,7):
        for j in range(1,i):
            f.write(chr(j+64+32))
        f.write('\n')
    f.close()
rat_lower_col()

# RAT - NAME - ROW
def rat_name_row():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====RAT NAME ROW====\n")
    for i in range(len(name)):
        for j in range(i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
rat_name_row()
```

```
def rat_name_col():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====RAT NAME COL====\n")
    for i in range(len(name)):
        for j in range(i+1):
            f.write(name[j])
        f.write('\n')
    f.close()
rat_name_col()
```

"-----"

LAT - NUM - ROW

```
def lat_num_row():
    f=open('patterns.txt', 'a+')
    f.write("====LAT NUM ROW====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
lat_num_row()
```

LAT - NUM - COL

```
def lat_num_col():
    f=open('patterns.txt', 'a+')
    f.write("====LAT NUM COL====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(str(k))
        f.write('\n')
    f.close()
lat_num_col()
```



```

# LAT - STAR
def lat_star():
    f=open('patterns.txt', 'a+')
    f.write("====LAT STAR====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write('*')
        f.write('\n')
    f.close()
lat_star()

# LAT - UPPER - ROW
def lat_upper_row():
    f=open('patterns.txt', 'a+')
    f.write("====LAT UPPER ROW====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(chr(i+64))
        f.write('\n')
    f.close()
lat_upper_row()

# LAT - UPPER - COL
def lat_upper_col():
    f=open('patterns.txt', 'a+')
    f.write("====LAT UPPER COL====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64))
        f.write('\n')
    f.close()
lat_upper_col()

```

```
# LAT - LOWER - ROW
def lat_lower_row():
    f=open('patterns.txt', 'a+')
    f.write("====LAT LOWER ROW====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(chr(i+64+32))
        f.write('\n')
    f.close()
lat_lower_row()
```

```
# LAT - LOWER - COL
def lat_lower_col():
    f=open('patterns.txt', 'a+')
    f.write("====LAT LOWER COL====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64+32))
        f.write('\n')
    f.close()
lat_lower_col()
```

```
# LAT - NAME - ROW
def lat_name_row():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====LAT NAME ROW====\n")
    for i in range(len(name)):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
lat_name_row()
```

```

# LAT - NAME - COL
def lat_name_col():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====LAT NAME COL====\n")
    for i in range(len(name)):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(i+1):
            f.write(name[k])
        f.write('\n')
    f.close()
lat_name_col()

"-----"

# IRAT - NUMBER - ROW
def irat_num_row():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT NUMBER ROW====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
irat_num_row()

# IRAT - NUMBER - COL
def irat_num_col():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT NUMBER COL====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(str(j))
        f.write('\n')
    f.close()
irat_num_col()

```



```
# IRAT - STAR
def irat_star():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT STAR====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write('*')
        f.write('\n')
    f.close()
irat_star()

# IRAT - UPPER - ROW
def irat_upper_row():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT UPPER ROW====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(chr(i+64))
        f.write('\n')
    f.close()
irat_upper_row()

# IRAT - UPPER - COL
def irat_upper_col():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT UPPER COL====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(chr(j+64))
        f.write('\n')
    f.close()
irat_upper_col()

# IRAT - LOWER - ROW
def irat_lower_row():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT LOWER ROW====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(chr(i+64+32))
        f.write('\n')
    f.close()
irat_lower_row()
```

File Edit Format Run Options Window Help

```
# IRAT - LOWER - COL
def irat_lower_col():
    f=open('patterns.txt', 'a+')
    f.write("====IRAT LOWER COL====\n")
    for i in range(6,0,-1):
        for j in range(0,i):
            f.write(chr(j+64+32))
        f.write('\n')
    f.close()
irat_lower_col()
```

```
# IRAT - NAME - ROW
def irat_name_row():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====IRAT NAME ROW====\n")
    for i in range(len(name)-1,-1,-1):
        for j in range(0,i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
irat_name_row()
```

```
# IRAT - NAME - COL
def irat_name_col():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====IRAT NAME COL====\n")
    for i in range(len(name),0,-1):
        for j in range(0,i):
            f.write(name[j])
        f.write('\n')
    f.close()
irat_name_col()
```

"-----"


```

# ILAT - NUM - ROW
def ilat_num_row():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT NUM ROW====\n")
    for i in range(5,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(str(i))
        f.write('\n')
    f.close()
ilat_num_row()

```

```

# ILAT - NUM - COL
def ilat_num_col():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT NUM COL====\n")
    for i in range(6,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(str(k))
        f.write('\n')
    f.close()
ilat_num_col()

```

```

# ILAT - STAR
def ilat_star():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT STAR====\n")
    for i in range(5,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for j in range(0,i):
            f.write('*')
        f.write('\n')
    f.close()
ilat_star()

```

```

# ILAT - UPPER - ROW
def ilat_upper_row():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT UPPER ROW====\n")
    for i in range(5,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(chr(i+64))
        f.write('\n')
    f.close()
ilat_upper_row()

```

```

# ILAT - UPPER - COL
def ilat_upper_col():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT UPPER COL====\n")
    for i in range(5,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64))
        f.write('\n')
    f.close()
ilat_upper_col()

```

```

# ILAT - LOWER - ROW
def ilat_lower_row():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT LOWER ROW====\n")
    for i in range(5,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(chr(i+64+32))
        f.write('\n')
    f.close()
ilat_lower_row()

```

```
# ILAT - LOWER - COL
def ilat_lower_col():
    f=open('patterns.txt', 'a+')
    f.write("====ILAT LOWER COL====\n")
    for i in range(5,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64+32))
        f.write('\n')
    f.close()
ilat_lower_col()
```

```
# ILAT - NAME - ROW
def ilat_name_row():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====ILAT NAME ROW====\n")
    for i in range(len(name)-1,-1,-1):
        for j in range(len(name)-1,i,-1):
            f.write(' ')
        for k in range(i+1):
            f.write(name[i])
        f.write('\n')
    f.close()
ilat_name_row()
```

```
# ILAT - NAME - COL
def ilat_name_col():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====ILAT NAME COL====\n")
    for i in range(len(name)-1,-1,-1):
        for j in range(len(name)-1,i,-1):
            f.write(' ')
        for k in range(i+1):
            f.write(name[k])
        f.write('\n')
    f.close()
ilat_name_col()
```

"-----"


```

# PRMD - NUM - ROW
def prmd_num_row():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD NUM ROW====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(str(i)+' ')
        f.write('\n')
    f.close()
prmd_num_row()

```

```

# PRMD - NUM - COL
def prmd_num_col():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD NUM COL====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(str(k)+' ')
        f.write('\n')
    f.close()
prmd_num_col()

```

```

# PRMD - STAR
def prmd_star():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD STAR====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write('*'+ ' ')
        f.write('\n')
    f.close()
prmd_star()

```

```

# PRMD - UPPER - ROW
def prmd_upper_row():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD UPPER ROW====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(chr(i+64)+' ')
        f.write('\n')
    f.close()
prmd_upper_row()

```

```

# PRMD - UPPER - COL
def prmd_upper_col():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD UPPER COL====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64)+' ')
        f.write('\n')
    f.close()
prmd_upper_col()

```

```

# PRMD - LOWER - ROW
def prmd_lower_row():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD LOWER ROW====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write(chr(i+64+32)+' ')
        f.write('\n')
    f.close()
prmd_lower_row()

```

```
# PRMD - LOWER - COL
def prmd_lower_col():
    f=open('patterns.txt', 'a+')
    f.write("====PRMD LOWER COL====\n")
    for i in range(1,6):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64+32)+' ')
        f.write('\n')
    f.close()
prmd_lower_col()
```

```
# PRMD - NAME - ROW
def prmd_name_row():
    name = 'bharath'
    f = open('patterns.txt', 'a+')
    f.write("====PRMD NAME ROW====\n")
    for i in range(1, len(name) + 1):
        for j in range(len(name), i, -1):
            f.write(' ')
        for k in range(i):
            f.write(name[i - 1] + ' ')
        f.write('\n')
    f.close()
prmd_name_row()
```

```
# PRMD - NAME - COL
def prmd_name_col():
    name = 'bharath'
    f = open('patterns.txt', 'a+')
    f.write("====PRMD NAME COL====\n")
    for i in range(1, len(name) + 1):
        for j in range(len(name), i, -1):
            f.write(' ')
        for k in range(i):
            f.write(name[k] + ' ')
        f.write('\n')
    f.close()
prmd_name_col()
```

"-----"


```

# IPRMD - NUM - ROW
def iprmd_num_row():
    f=open('patterns.txt', 'a+')
    f.write("====IPRMD NUM ROW====\n")
    for i in range(6,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(str(i)+' ')
        f.write('\n')
    f.close()
iprmd_num_row()

```

```

# IPRMD - NUM - COL
def iprmd_num_col():
    f=open('patterns.txt', 'a+')
    f.write("====IPRMD NUM COL====\n")
    for i in range(6,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(str(k)+' ')
        f.write('\n')
    f.close()
iprmd_num_col()

```

```

# IPRMD - STAR
def iprmd_star():
    f=open('patterns.txt', 'a+')
    f.write("====IPRMD STAR====\n")
    for i in range(6,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(0,i):
            f.write('*'+ ' ')
        f.write('\n')
    f.close()
iprmd_star()

```

```
# IPRMD - UPPER - ROW
```

```
def iprmd_upper_row():  
    f=open('patterns.txt', 'a+')  
    f.write("====IPRMD UPPER ROW====\n")  
    for i in range(6,0,-1):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(chr(i+64)+' ')  
        f.write('\n')  
    f.close()  
iprmd_upper_row()
```

```
# IPRMD - UPPER - COL
```

```
def iprmd_upper_col():  
    f=open('patterns.txt', 'a+')  
    f.write("====IPRMD UPPER COL====\n")  
    for i in range(6,0,-1):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(1,i):  
            f.write(chr(k+64)+' ')  
        f.write('\n')  
    f.close()  
iprmd_upper_col()
```

```
# IPRMD - LOWER - ROW
```

```
def iprmd_lower_row():  
    f=open('patterns.txt', 'a+')  
    f.write("====IPRMD LOWER ROW====\n")  
    for i in range(6,0,-1):  
        for j in range(6,i,-1):  
            f.write(' ')  
        for k in range(0,i):  
            f.write(chr(i+64+32)+' ')  
        f.write('\n')  
    f.close()  
iprmd_lower_row()
```

```

# IPRMD - LOWER - COL
def iprmd_lower_col():
    f=open('patterns.txt', 'a+')
    f.write("====IPRMD LOWER COL====\n")
    for i in range(6,0,-1):
        for j in range(6,i,-1):
            f.write(' ')
        for k in range(1,i):
            f.write(chr(k+64+32)+' ')
        f.write('\n')
    f.close()
iprmd_lower_col()

```

```

# IPRMD - NAME - ROW
def iprmd_name_row():
    name='bharath'
    f=open('patterns.txt', 'a+')
    f.write("====IPRMD NAME ROW====\n")
    for i in range(len(name),0,-1):
        for j in range(len(name),i,-1):
            f.write(' ')
        for k in range(i):
            f.write(name[i-1]+' ')
        f.write('\n')
    f.close()
iprmd_name_row()

```

```

# IPRMD - NAME - COL
def iprmd_name_col():
    name = 'bharath'
    f = open('patterns.txt', 'a+')
    f.write("====IPRMD NAME COL====\n")
    for i in range(len(name), 0, -1):
        for j in range(len(name), i, -1):
            f.write(' ')
        for k in range(i):
            f.write(name[k] + ' ')
        f.write('\n')
    f.close()
iprmd_name_col()

```


=====RAT NUMBER ROW=====

1

22

333

4444

55555

=====RAT NUMBER COL=====

0

01

012

0123

01234

=====RAT STAR=====

*

**

=====RAT UPPER ROW=====

A

BB

CCC

DDDD

EEEEEE

=====RAT UPPER COL=====

A

AB

ABC

ABCD

ABCDE

=====RAT LOWER ROW=====

a

bb

ccc

dddd

eeeee

=====RAT LOWER COL=====

a

ab

abc

abcd

abcde

=====RAT NAME ROW=====

b

hh

aaa

rrrr

aaaaa

tttttt

hhhhhhh

=====RAT NAME COL=====

b

bh

bha

bhar

bhara

bharat

bharath

=====LAT NUM ROW=====

1

22

333

4444

55555

=====LAT NUM COL=====

1

12

123

1234

=====LAT STAR=====

*

**

=====LAT UPPER ROW=====

A

BB

CCC

DDDD

EEEE

=====LAT UPPER COL=====

A

AB

ABC

ABCD

=====LAT LOWER ROW=====

a

bb

ccc

dddd

eeee

=====LAT LOWER COL=====

a

ab

abc

abcd

=====LAT NAME ROW=====

b

hh

aaa

rrrr

aaaaa

tttttt

hhhhhhh

=====LAT NAME COL=====

b

bh

bha

bhar

bhara

bharat

bharath

=====IRAT NUMBER ROW=====

666666

55555

4444

333

22

1

=====IRAT NUMBER COL=====

012345

01234

0123

012

01

0

=====IRAT STAR=====

**

*

=====IRAT UPPER ROW=====

FFFFFF

EEEEEE

DDDD

CCC

BB

A

=====IRAT UPPER COL=====

ABCDE

ABCD

ABC

AB

A

=====IRAT LOWER ROW=====

ffffff

eeeeee

dddd

ccc

bb

a

=====IRAT LOWER COL=====

abcde

abcd

abc

ab

a

=====IRAT NAME ROW=====

hhhhhhh

tttttt

aaaaa

rrrr

aaa

hh

b

=====IRAT NAME COL=====

bharath

bharat

bhara

bhar

bha

bh

b

=====ILAT NUM ROW=====

55555

4444

333

22

1

=====ILAT NUM COL=====

12345

1234

123

12

1

=====ILAT STAR=====

**

*

=====ILAT UPPER ROW=====

EEEE

DDDD

CCC

BB

A

=====ILAT UPPER COL=====

ABCD

ABC

AB

A

=====ILAT LOWER ROW=====

eeee

dddd

ccc

bb

a

=====ILAT LOWER COL=====

abcd

abc

ab

a

=====ILAT NAME ROW=====

hhhhhhh

tttttt

aaaaa

rrrr

aaa

hh

b

=====ILAT NAME COL=====

bharath

bharat

bhara

bhar

bha

bh

b

=====PRMD NUM ROW=====

1

2 2

3 3 3

4 4 4 4

5 5 5 5 5

=====PRMD NUM COL=====

0

0 1

0 1 2

0 1 2 3

0 1 2 3 4

=====PRMD STAR=====

*

* *

* * *

* * * *

* * * * *

=====PRMD UPPER ROW=====

A

B B

C C C

D D D D

E E E E E

=====PRMD UPPER COL=====

A
A B
A B C
A B C D

=====PRMD LOWER ROW=====

a
b b
c c c
d d d d
e e e e e

=====PRMD LOWER COL=====

a
a b
a b c
a b c d

=====PRMD NAME ROW=====

b
h h
a a a
r r r r
a a a a a
t t t t t t
h h h h h h h

=====PRMD NAME COL=====

b
b h
b h a
b h a r
b h a r a
b h a r a t
b h a r a t h

=====IPRMD NUM ROW=====

6 6 6 6 6
5 5 5 5
4 4 4
3 3
2

=====IPRMD NUM COL=====

1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

=====IPRMD STAR=====

* * * * *

* * * * *

* * * *

* * *

* *

*

=====IPRMD UPPER ROW=====

F F F F F F

E E E E E

D D D D

C C C

B B

A

=====IPRMD UPPER COL=====

A B C D E

A B C D

A B C

A B

A

=====IPRMD LOWER ROW=====

f f f f f f

e e e e e

d d d d

c c c

b b

a

=====IPRMD LOWER COL=====

a b c d e

a b c d

a b c

a b

a

=====IPRMD NAME ROW=====

h h h h h h h

t t t t t t t

a a a a a

r r r r r

a a a

h h

b

=====IPRMD NAME COL=====

b h a r a t h

b h a r a t

b h a r a

b h a r

b h a

b h

b